

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

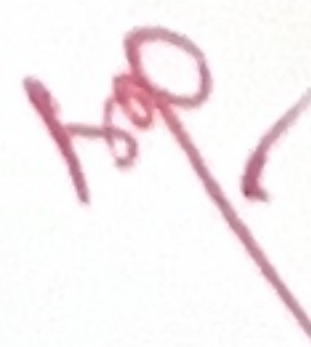
Code of Conduct:

I 19256588 (Rev) 1904V (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:





ASSESSMENT

Tool

1

Course : Artificial Intelligence

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Answers:

1 8-Puzzle Problem:

State representation:

It is represented by 3×3 grid containing 1 To 8 and one blank space.

Initial State:

The starting arrangement of the tiles given to Puzzle.

eg

1	2	3
4	□	6
7	5	8

Goal Test:

The goal is reached when all tiles are arranged with Blank Space in last position

eg:

1	2	3
4	5	6
7	8	□

Possible Action:

- up
- Down
- Left
- Right.

- The Blank tile can move four possible direction
- The maximum branching factor is 4.

Corner: 2 moves

Edges: 3 moves

Center: 4 moves.

2 AI-Agent for Hospital patient Triage:

Performance Measure:

- Correctly identifying patient severity.
- Reducing waiting time for critical patients.
- Fast response to incoming patients.
- Improving patients safety and treatment efficiency.

Rationality:

A Rational Agent select the Action

that gives the best expected outcome based on the info.

When is used?

- Uses the patient's symptoms and medical data correctly.
- Gives higher priority to emergency cases.
- Follow hospital Triage guidelines.
- Adapts its decision when new patient information becomes available.

3 AI Agent for Warehouse Robots!

Justification:

- Learn the shortest and most efficient route
- Avoid obstacles and prevent collisions.
- Adapt To changes in Warehouse layout
- Optimize battery usage and Task scheduling.

Why NOT other Agent Types:

- Simple Reflex Agent : Works only for fixed situation and cannot adapt.

- Model - Based Agent : Maintain an Internal model but has limited Capability.

- Goal Based Agent : Reaches goal but may not choose the most efficient path.

- Learning Agent : Continuously improves Performance, making it ideal for dynamic Warehouse environments.

Conclusion :

A Learning Agent is the best choice because Warehouse environment change frequently, and robot must continuously learn and optimize their navigation, Picking, and delivery task for high efficiency.